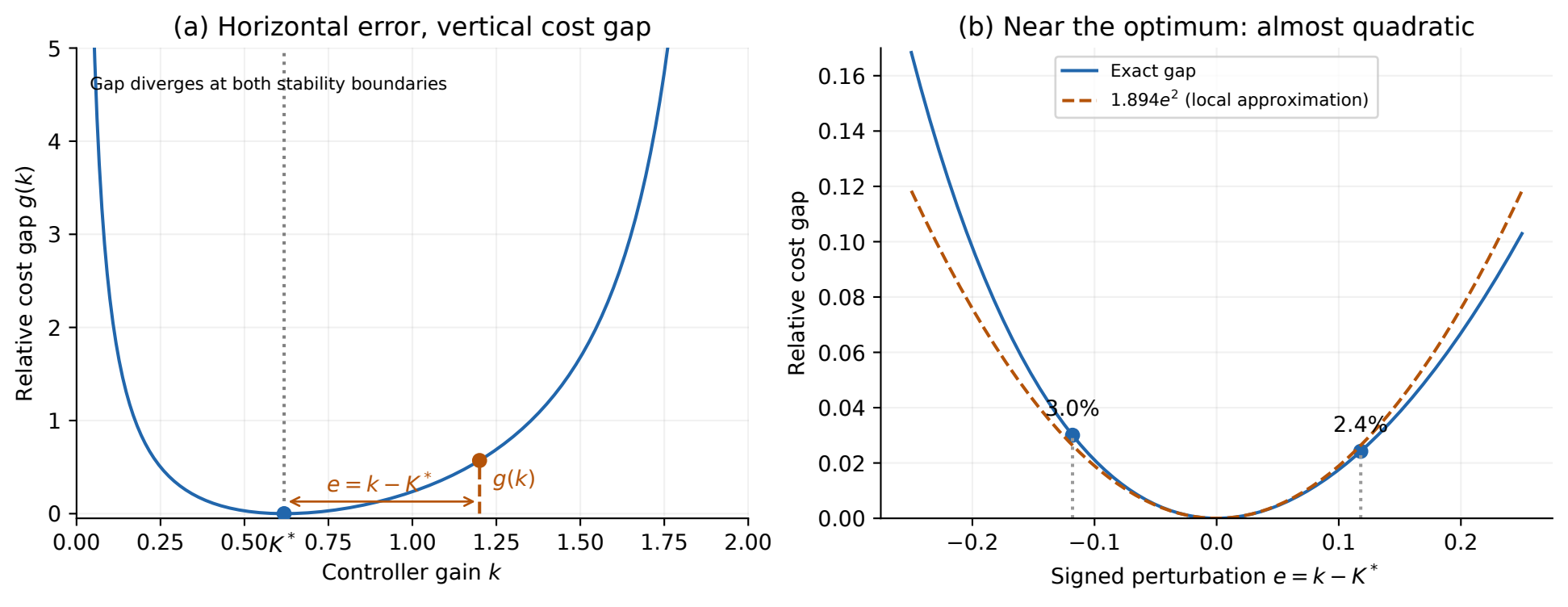
(a) Horizontal error, vertical cost gap
Gap diverges at both stability boundaries
Relative cost gap $g(k)$
Controller gain k
K^*
$e = k - K^*$
$g(k)$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
0
1
2
3
4
5
(b) Near the optimum: almost quadratic
Exact gap
$1.894e^2$ (local approximation)
3.0%
2.4%
Relative cost gap
Signed perturbation $e = k - K^*$
−0.2
−0.1
0.0
0.1
0.2
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16